January 2026 CSE 220
Online on Signals and Their Properties
Subsections:
Total Marks: 10
Total Time: 40 Minutes

In this assignment, you need to implement time scaling and shifting of a sampled continuous-time signal in Python. You also need to plot the graphs of each of these signals.

Let the base signal be $x(t)$ (provided in the template file). You will compute and plot:

$x(t),\ y(t) = x(\alpha t + \beta)$; Where $\alpha > 0$

Tasks

★ Generate the time axis t and compute $x(t)$. (given)

★ Implement a function *transform_signal(t, x, alpha, beta)* that produces $y(t) = x(\alpha t + \beta)$

★ Assume the signal is bounded between -pi and pi, values outside the range are equal to 0.

- If $\alpha > 1$, the signal will be compressed. For example: for α **= 2**; $y(0) = x(0)$, $y(1) = x(2)$, $y(2) = x(4)$ and so on.
- If $\alpha < 1$, the signal will be expanded and you may require new samples between original values. For example: for α **= 0.5**; $y(0) = x(0)$, $y(2) = x(1)$ but $y(1) = x(0.5)$ which is not available. So, you have to interpolate the value. You will implement a function *interpolate_signal(...)* to interpolate the missing sample values. Missing values will be calculated from the average of the nearest left and right values. $y(1) = 0.5 * (x(0) + x(1))$.
- In case of time shift, the signal shifts to left or right. For example: for β **= 1**; $y(0) = x(-1)$, $y(1) = x(0)$ and so on.

★ Plot (with proper labels and legend) on the same figure: $x(t),\ y(t)$.

★ In the main function, make an infinite loop that will take alpha and beta repeatedly, and show plots on each iteration. The program will end upon receiving ‘q’.

Marks Distribution

- Implementing scaling: 6
- Implementing shifting: 2
- Completing the loop: 2